

Saba Kareem

Islamabad, Pakistan 44000

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Professional Summary

Ambitious researcher with dual Master's degrees and a strong foundation in Artificial Intelligence (AI), Big Data Analytics, and Healthcare Informatics. Specialized in Federated Learning (FL) and Spiking Neural Networks (SNNs) with a focus on privacy-preserving healthcare solutions. Proven track record of leading research projects, securing academic achievements, and publishing in reputable journals. Passionate about advancing healthcare technologies through data-driven innovations, making significant contributions to both academic and applied research environments.

Education

Master of Science in Information Technology (Research Track)

Universiti Teknologi Petronas (UTP), Perak, Malaysia

2016 – 2019

Focus: Machine Learning, Big Data Analytics in Healthcare

Thesis: Big Data Frameworks for Crisis Management in Healthcare Systems

Roles: Vice President, AISPC; UTP Ambassador

Master of Science in Information Technology

Quaid-i-Azam University, Islamabad, Pakistan

2010 – 2012

Focus: Agile Methodologies, Scrum Project Management, Programming (C, C++, SQL)

Role: President, Information Technology Society

Bachelor of Science in Computer Science and Programming

University of The Punjab, Islamabad, Pakistan

2007 – 2009

Research Interests

- Federated Learning and its Applications in Healthcare
- Spiking Neural Networks (SNNs) for Real-Time Health Data Analysis
- Privacy-Preserving Machine Learning Techniques
- Big Data Analytics for Crisis Management and Public Health Policy

Professional Experience

Graduate Research Assistant

Universiti Teknologi Petronas (UTP), Perak, Malaysia

2016 – 2019

- Spearheaded research on privacy-preserving big data analytics in healthcare.
- Developed machine learning models for healthcare crisis management and predictive analytics.
- Mentored students in AI-based research projects and supervised data analysis tasks.
- Presented research findings at international conferences, enhancing visibility of healthcare AI innovations.

Junior Data Engineer

WebBeds, Dubai

2012 – 2014

- Designed and implemented data pipelines for scalable healthcare data solutions.
- Optimized big data workflows to support real-time healthcare analytics.
- Collaborated with cross-functional teams to enhance the efficiency of data management systems.

Technical Skills

- **Programming:** Python, C, C++, SQL
- **Frameworks & Tools:** Hadoop, TensorFlow, PyTorch, BindsNET, Federated Averaging Algorithms
- **Data Analytics:** Association Rule Mining, Big Data Frameworks, Data Visualization
- **Project Management:** Agile, Scrum Methodologies
- **Neural Network Tools:** NeuCom, NeuCube, Dynamic Evolving Neural-Fuzzy Inference System (DENFIS)

Certifications

- IELTS (Proficient)
- Research Methodology Certification
- Brain-Inspired Artificial Intelligence (Spiking Neural Networks Focus)
- Advanced Data Analytics in Healthcare (2023)
- Short Course on Neural Networks and Brain-Inspired AI with Prof. Nikola Kasabov (Focused on NeuCom, NeuCube, DENFIS, and related methodologies)

Selected Publications

1. Saba Kareem, Rohiza Binti Ahmad, Aliza Binti Sarlan (2017). *Framework for the Identification of Fraudulent Health Insurance Claims using Association Rule Mining*, IEEE Conference on Big Data and Analytics (ICBDA).
2. Saba Kareem, Rohiza Binti Ahmad, Aliza Binti Sarlan (2018). *Extraction of Associations from Electronic Health Records using Association Rules Mining*, Journal of Advanced Research in Dynamical and Control Systems.
3. Saba Kareem, Ommar Hayat Abbasi (Under Review). *An Automated Approach for Crisis Management: A Case Study of Big Data Analytics in Healthcare*.
4. Saba Kareem, Nurul Hashimah Binti Ahamed Hassain Malim (Under Review). *Neuromarketing and Neuroscience: Techniques, Applications, and Challenges*.

Professional Affiliations

- Vice President, AISPC, Universiti Teknologi Petronas
- UTP Ambassador (Microsoft Imagine Cup Participant)
- President, Information Technology Society, Quaid-i-Azam University
- Member, English Debate Society

Languages

- Arabic (Proficient)
- English (Fluent)
- Malay (Basic)
- Urdu (Native)